2

History of the Automobile (or How We Learned to Get Off Our High Horse)

The automobile ranks among the most important inventions in human history. Since its inception, the passenger car has provided unprecedented levels of autonomy, allowing greater freedom in where we live, where we work, and where we travel. Today, there are over one billion cars on the road, and more than 99% are powered by ICEs [1]. But, if you haven't been living under a rock for the past few years (or even if you have), you've probably heard the idea that the ICE has reached its final days and will soon be replaced by the new kid in town: the electric vehicle (EV).

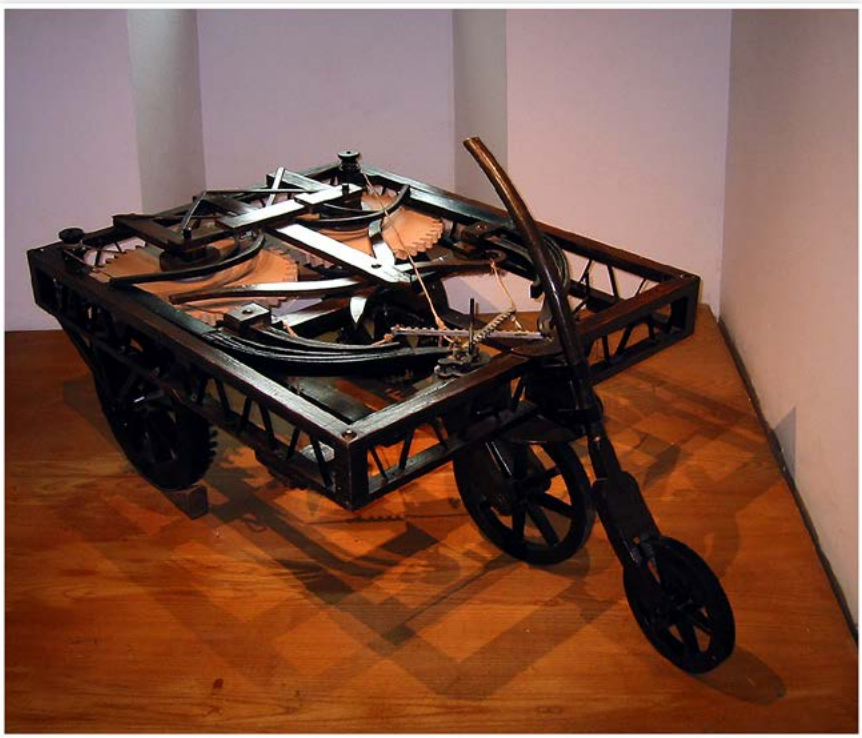
Except—electric cars aren't all that new. In fact, they've been around for a long time. And we're not talking about GM's EV1 of the 1990s. Or even the smattering of EV prototypes built in the 1960s and 1970s. No, we're talking about the very first EVs, invented way back in the 19th century. The electric car and the gasoline car are nearly the same age! So, why does the misconception that EVs are a new technology persist? And why is it that the vast majority of cars today are powered by ICEs? To answer these questions, we need to take a look back in time to see how the automobile first came to be and how we ended up where we are today.

The Very Beginning

The idea of a self-propelled vehicle dates far back in history—too far to determine who first conceived of it. Around the 8th century BC, Homer wrote in the *Iliad* about a "self-moved" vehicle that was "obedient to the beck of gods" [2], but the idea likely originated long before then. Some might even say we have "documentary evidence" of stone-aged, foot-powered vehicles from the well-known cartoon *The Flintstones*. Luckily, we've come a long way since the days of Fred and Barney.

Fast forward to the 15th century when one of the most famed inventors in history, Leonardo da Vinci, sketched out plans for a self-propelled cart. The cart runs on clockwork and is wound up by rotating the wheels in the opposite direction of intended travel, much like a wind-up toy car [3]. Historians believe that the cart was not intended as a passenger vehicle, especially since it lacks any type of seat, and was instead designed to be a novelty, perhaps at a Renaissance festival [4]. While da Vinci never built his self-propelled cart, several modern scholars have attempted to create it. However, da Vinci's sketches were unclear and confusion over exactly how the vehicle was powered caused most efforts to fail. That changed in 2004 when a team of engineers led by Paolo Galluzzi, director of the Institute and Museum of Science in Florence, Italy, successfully created a prototype of da Vinci's vehicle, proving once and for all that it worked [4, 5].

FIGURE 2.1 Reconstruction of da Vinci's self-propelled cart on display at Château du Clos Lucé in Amboise, France.



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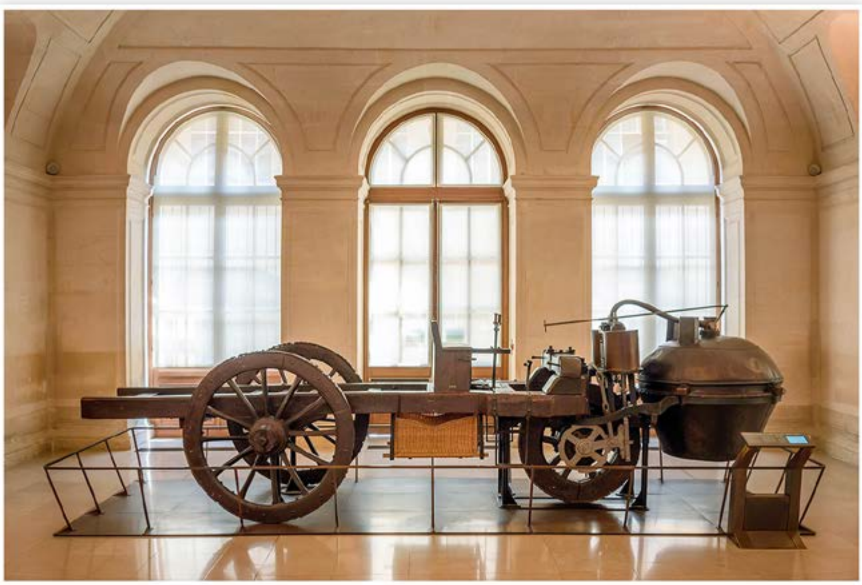
Raising Steam

Although da Vinci conceived of his self-propelled cart hundreds of years earlier, the first viable self-propelled vehicles *actually produced* were powered by steam. Steam engines are ECEs. That's not a typo—they are *external* combustion engines! It's beyond the scope of this book to go into detail on the inner (or external) workings of a steam engine, but we'll give you the short version. Very briefly, steam engines usually work by burning coal to heat up water and create steam, which is then used to propel the vehicle. If you're interested in learning more, we recommend checking out *Steam Traction on the Road* by Anthony Burton [6].

With this technology, French engineer Nicolas-Joseph Cugnot built what can be considered the world's first true automobile in 1769 [7, 8]. The vehicle was a three-wheeled military tractor powered by a steam engine and designed to haul artillery. It could carry up to four passengers and reached a thrilling speed of just over 2 miles per hour (mph). The vehicle could run for 10-20 minutes before it needed to stop and build up enough steam to run again. Unfortunately, the design of Cugnot's vehicle made it susceptible to tipping over when it wasn't towing military equipment. And on the tractor's first trip around Paris in 1770, it crashed into a wall (although damage was minimal given the vehicle's low speed) [9].

FIGURE 2.2 Nicolas-Joseph Cugnot's steam-powered tractor, considered to be the first true automobile.



Reprinted from https://commons.wikimedia.org/wiki/File:Joseph_Cugnot%27s_1770_Fardier_%C3%A0_Vapeur,_Mus%C3%A9e_des_arts_et_m%C3%A9tiers,_Paris_2015.jpg Licensed under under the Creative Commons CC0 1.0 Universal Public Domain Dedication.

Despite the limitations of Cugnot's design, subsequent steam vehicles started to emerge in the following years. By 1800, steam buses were running routes in Paris [2]. In 1801, British inventor Richard Trevithick debuted his passenger steam vehicle, nicknamed the "Puffer" or the "Puffing Devil" because of the steam that billowed out into the air. On Christmas Eve of that year, he took several passengers on a short ride, with one of them noting the vehicle "went off like a little bird … going faster than I could walk" [10]. Unfortunately, a few days later, the engine caught fire, and the short-lived Puffer was destroyed [11].

Other inventors around Europe and in the United States built and demonstrated various steam carriages throughout the 19th century. One inventor of note was Francis Curtis, who lived in Newburyport, Massachusetts. In the mid-1800s, Curtis built a steam carriage for a customer, who then fell behind on his payments. In response, Curtis took the carriage back, in the first recorded instance of a vehicle repossession [2]. Apparently, Curtis also caused quite the ruckus in his neighborhood by driving around in his steam carriage, which greatly irritated his neighbors. This led to a warrant being issued for his arrest, likely for disturbing the peace, although the actual charge has been lost to history. When the sheriff arrived, Curtis was involved in yet another first when he took off in his steam carriage with the officer in pursuit on foot, turning his carriage into the first getaway car in American history [12, 13].

In Britain, the heyday of steam coaches came in the 1830s, when a variety of routes were run, including one between London and Cambridge [2]. Steam-propelled vehicles faced a number of obstacles, however, including public disapproval and fierce lobbying from the horse-drawn carriage industry. In response, Britain implemented the Locomotives on Highways Act of 1865, also known as the Red Flag Act [14]. One of the more absurd traffic laws in British history, the legislation stipulated that all horseless carriages on the road had to be preceded by a person on foot carrying a red flag. Another of the act's provisions set the speed limit for horseless carriages at 2 mph within cities and 4 mph on country roads. Horse-drawn carts could achieve speeds of about 10–15 mph, so implementing such a low speed limit for horseless carriages was a not-so-subtle attempt to render them useless. The law effectively damped innovation in road transport in Britain until it was repealed in 1896 [15, 16].

Although interest in steam vehicles began to decline toward the turn of the century, light steam cars were still being built in the United States, France, Germany, and Denmark. In America, the Stanley brothers represented the pinnacle of steam-car design. They debuted their first steam-powered car, the Stanley Steamer, in 1898 at the Charles River Park velodrome in Massachusetts. They entered their horseless carriage in competitions of speed and endurance and placed well in both [17]. By 1899, they had sold more than 200 of their steam-powered cars, which made them the most successful automaker in the United States at the time [18]. In 1906, the Stanley Rocket, another steam-powered car built by the Stanley brothers, set the world record for speed. With an aerodynamic design that resembled an upside-down canoe, the Rocket reached a speed of 127.66 mph—the fastest steam vehicle until 2009 (although a gasoline car beat the Rocket's record in 1910) [19]. The Stanley cars of the early 20th century rivaled both the performance and the sales of their gasoline-powered contemporaries [18].

FIGURE 2.3 1914 Stanley Steamer.

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But while the Stanley Steamers are perhaps the most famous steam automobiles of the early 20th century, Stanley was not the largest producer of steamers at the time. That honor goes to a company that was known for a different product: the White Sewing Machine Company. Founded by Thomas H. White in 1858 and incorporated in 1876 [20], the company was a successful manufacturer of sewing machines based in Cleveland, Ohio. White's sons were interested in the automotive business, however, and they began making steam cars after one of them, Rollin White, invented a semi-flash boiler in the late 1890s. By 1900, they had built four White steam cars and the White Sewing Machine Company added automobiles to its portfolio. White steamers gained prominence after setting a world speed record in 1905, and in 1906, the company sold 1,534 cars—about double the number of Stanley Steamers sold [21].

In 1906, White's automobile business separated from the White Sewing Machine Company and was renamed the White Company. A White Company steamer was among the first official White House automobile fleet established in 1909 by President William Howard Taft, further cementing the White steam car's prodigious reputation. Unlike Stanley, however, White did not exclusively produce steam cars, and they added a gasoline model to their lineup in 1910. One year later, the last White steam car was built, and in 1918, the White Motor Company ceased manufacturing passenger cars to focus instead on their line of trucks [22].

FIGURE 2.4 President William Howard Taft's White Motor Company steam car in 1909.

Library of Congress, Prints & Photographs Division, George Grantham Bain Collection, [LC-DIG-ggbain-03212]

Electric Shock

Many think of electric cars as a modern, even futuristic technology—and today's EVs are indeed frequently equipped with advanced software and other high-tech features—but electric cars have been around since the 19th century. Determining when the first electric car was built is tricky, since it depends on your definition of an electric car. In 1828, Hungarian engineer Ányos Jedlik made a small-scale model, but some historians don't count his invention as the first EV because of its size [23, 24]. Instead, Thomas Davenport, an American, is sometimes credited as the inventor of the electric car. In 1835, he built a small locomotive powered by two electromagnets, a pivot, and a battery [24, 25]. Around the same time, somewhere between 1832 and 1839 (the exact date is not known), Scotland's Robert Anderson built a motorized carriage—another potential contender for the first EV.

Regardless of whose vehicle was the first true electric car, these early iterations all suffered from the same technological shortcoming: Batteries at the time were not yet rechargeable, so these EVs were more for show than practical transportation [26].

That changed in 1859 when French physicist Gaston Planté invented the rechargeable lead-acid storage battery [27] and paved the way for more practical EVs. One such EV was built in 1884 by Thomas Parker, who is another individual sometimes credited with the invention of the electric car. That is not Parker's only claim to fame, though—Parker is also responsible for electrifying the London underground [25].

In the United States, William Morrison of Des Moines, Iowa, is often cited as the creator of the first successful and practical American electric automobile. His vehicle, which could reach speeds of 15-20 mph, had a four-horsepower motor and a 24-cell battery that needed to recharge every 50 miles [26, 28]. Morrison exhibited his electric carriage at the 1893 Chicago World's Fair (also known as the World's Columbian Exposition), causing quite the stir and inspiring other inventors.

The next few years saw many successes for the EV. New York City commissioned a fleet of electric taxis [23]. The Pope Manufacturing Company of Connecticut became the first major manufacturer of commercially successful electric cars. An electric car defeated five gasoline-powered vehicles in a race in 1896 [28]. By 1900, there were around 12 manufacturers of EVs in the United States, and electric cars accounted for about one-third of cars on the road [24, 28].

The Heat of Combustion

That brings us to the technology that has dominated propulsion systems for a century: the internal combustion engine.

The answer to who invented the internal combustion engine is murky, but a solid contender is Franco-Swiss inventor François Isaac de Rivaz. In 1807, he built and patented a hydrogen-powered combustion engine. Shortly after, he apparently built a primitive automobile for his engine to power—perhaps the very first ICE-powered car [29].

However, de Rivaz's engine was never commercially successful—Belgian engineer Jean Joseph Étienne Lenoir holds the distinction of inventing the first practical, commercially viable internal combustion engine sometime around 1858–1859. His two-stroke, single-cylinder engine resembled a horizontal double-acting steam beam engine and ran on a mixture of coal gas and air [30]. Notably, his engine was "compression-less" and thus did not compress the air-fuel mixture before ignition, which is standard practice in modern ICEs. Lenoir also patented a spark plug, the design of which is essentially the same as the spark plugs used today [31]. Although Lenoir's engine was noisy and achieved only about 4% efficiency, it ran smoothly and was incredibly durable [32]. By 1865, more than 1,400 of his engines were in use in France and Britain for applications requiring low power, such as pumping and printing [32].

Lenoir is also sometimes credited as creating the first vehicle powered by an ICE. During 1862–1863, he affixed an improved version of his engine to a carriage to create the "Hippomobile." However, Lenoir's engine was too small to efficiently power the heavy vehicle—the engine could only reach 100 revolutions per minute, resulting in a top speed of less than 4 mph [33].

Still, Lenoir's engine inspired a household name in internal combustion: Nikolaus Otto. Otto learned of Lenoir's engine while working as a traveling salesman in Cologne, and in 1861, he created an experimental replica of the engine [34, 35]. A few years later, he formed a partnership with industrialist Eugen Langen to build and sell the engine. The duo debuted their first design, a noisy two-stroke engine, which differed from Lenoir's design by compressing the gas before ignition, in 1867 at the Paris Exhibition. The engine was far more fuel efficient than the other engines at the exhibition, and Otto and Langen took first prize. The engine was also good enough to achieve commercial success, and so they founded Gasmotorenfabrik Deutz AG It was at this time that Otto and Langen hired Gottlieb Daimler and Wilhelm Maybach (two other well-known names) for their engineering team [34].

Despite the success of Otto's two-stroke engine, he felt it was limited in potential power generation. He shifted his focus to creating a four-stroke engine based on the thermodynamic cycle that he devised (today called the Otto cycle). It took many years of development, but in 1876, success was achieved. The principle of his engine remains the basis of all four-stroke internal combustion engines today. The four-stroke cycle goes as follows:

1. **Intake**. Air (and sometimes fuel as well) is drawn into the cylinder as the piston moves downward.
2. **Compression**. The piston moves upward, and the air (or air-fuel mixture) is compressed. In direct injection gasoline and diesel engines, the fuel is injected either in the intake stroke or during the compression stroke before ignition.
3. **Ignition**. Near the top of the compression stroke, the mixture is ignited, driving the piston downward and producing power. How the mixture is ignited forms the main difference between spark ignition (gasoline) and compression ignition (diesel) engines.
4. **Exhaust**. The piston moves upward again, pushing the exhaust gases out of the cylinder.

Otto's four-stroke engine was a huge success, and more than 50,000 were sold over the next couple of decades [35]. We should note, however, that Otto did not invent the principle for the four-stroke engine. Several other individuals, including Christian Reithmann and Alphonse Beau de Rochas, had previously described the process in the 1860s. Despite this, Otto is still recognized as the creator of the first successful four-stroke engine.

While Otto built the progenitor of modern gasoline (spark ignition) ICEs, his engines were designed for stationary applications. In fact, Otto had no real interest in transportation. Daimler, who had been working at Deutz AG for several years, was of a different mind. He thought Otto didn't recognize the full potential of the engine, and he left Deutz AG after the two had a falling out. Daimler and Maybach set up shop together in Stuttgart, Germany in 1882 and began working on their shared goal: creating combustion engines for vehicles [36].

By the end of the next year, they had developed a small, high-speed ICE with "hot-tube" ignition, which consisted of an incandescent tube in the cylinder head [36, 37]. They affixed this engine to a wooden bicycle, effectively creating the world's first motorcycle. Daimler's next engine, dubbed the "grandfather clock" because of its appearance, was used to power a motor carriage, becoming one of the earliest gasoline-powered automobiles [33].

FIGURE 2.5 A replica of the first motorcycle made by Daimler and Maybach.

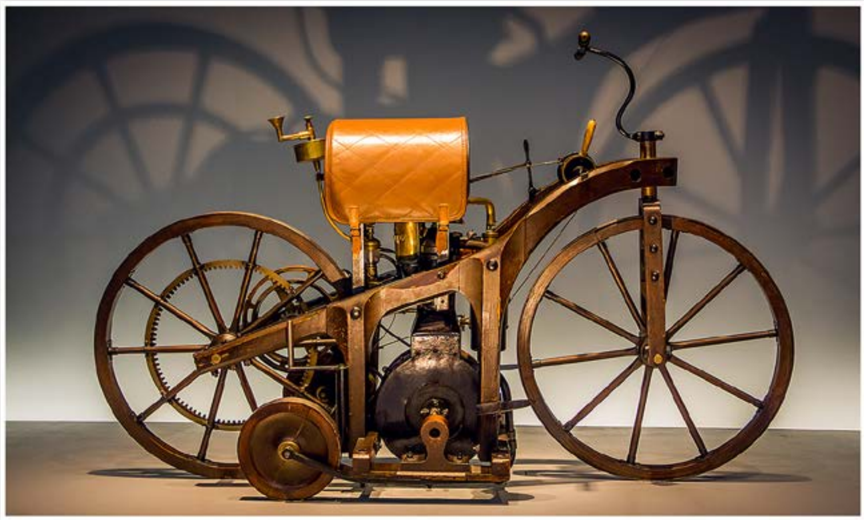

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FIGURE 2.6 1896 Daimler Riemenwagen with an engine designed by Maybach.

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Daimler was early in the combustion mobility game, but his contemporary and competitor Karl Benz is frequently credited with building the world's first practical ICE-powered automobile. Benz was a German engineer who developed internal combustion engines and, in 1885, created a motorized tricycle. The tricycle featured several other of Benz's inventions, including an electric ignition system, a carburetor, and a water-cooling system [38]. To promote his tricycle, Benz and his wife Bertha held a public demonstration. The tricycle reached a speed of 8 mph, but the demo ended with Benz accidentally running into a brick wall (running into walls seems to be a theme for early drivers, doesn't it?) [38, 39].

Benz started commercially selling his three-wheeled automobile in 1886, but business at the beginning was slow. Now, Bertha Benz was a fervent believer in her husband's dream and had already helped support his ventures financially [40]. She also had a knack for marketing and cooked up a scheme to help sell Benz's vehicles. In August of 1888, she took Benz's automobile without telling him and, along with her two teenage sons, embarked on a journey of more than 50 miles from Mannheim to Pforzheim, the first long-distance road trip in an automobile. The journey was not without a few hiccups—Bertha had to repair several technical malfunctions herself and her sons had to push the car up the bigger hills—but after 12 hours, they arrived safely at their destination [40]. The stunt successfully cultivated interest in Benz's motor cars and also led Benz to make some improvements to his vehicle, including adding gears for driving uphill and incorporating brake pads [39].

Sales picked up quickly for Benz, and by the end of the year he had employed 50 workers to manufacture his tricycle car. Benz continued to make improvements to his design, including securing a patent for the double-pivot steering system and building a four-wheeled automobile, both in the early 1890s. Benz also holds the distinction of being the first person to obtain a driver's license, years before they would become mandatory [33].

FIGURE 2.7 1886 Benz Patent-Motorwagen—the first car by Karl Benz.

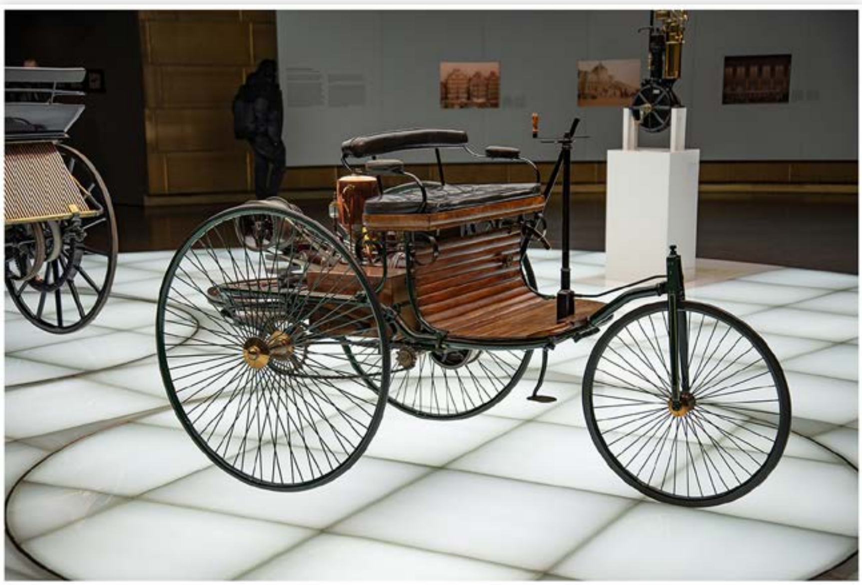

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FIGURE 2.8 1895 Benz Omnibus motor carriage.

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As Daimler and Benz focused on vehicles powered by spark-ignition engines, Rudolf Diesel was devising another type of technology: the diesel (or compression ignition) engine. From an early age, Diesel was fascinated by thermodynamics, and while he worked in refrigeration as a young man, his passion lay in engines. After learning about the Carnot cycle—a theoretical thermodynamic cycle that describes the maximum possible efficiency a heat engine can achieve during the conversion of heat into work—Diesel set about to make a more practical thermodynamic cycle based on the principle: the Diesel cycle.[1]

The engine Diesel designed employed a different ignition method than the gasoline engines at the time. Instead of compressing a mixture of fuel and air and triggering ignition with a spark plug, Diesel's engine compressed only air until its temperature was hot enough to spontaneously (or close enough) ignite the fuel upon injection [41]. Diesel patented this engine in 1892, but at that point, the engine was merely theoretical and existed only on paper. After receiving his patent, Diesel began the arduous process of building his engine. He partnered with Maschinenfabrik Augsburg and Friedrich Krupp AG, but struggled to achieve reliable combustion in his early experiments. Compression ignition engines require very high in-cylinder pressures to achieve high enough temperatures to ignite the fuel. Additionally, while Diesel envisioned his engine as being capable of running on a multitude of fuels, he discovered that finding a fuel that would actually ignite when injected into the

[1] It's important to note that real internal combustion engines are not thermodynamic cycles, and their efficiency is not actually limited by Carnot or any closed thermodynamic cycle. The three key differences are that real engines are (1) not closed systems, (2) use air-fuel mixtures, not a single working fluid, and (3) are not steady flow (they are pulsating flow). As a result, the Otto and Diesel cycles are at best approximations of gasoline and diesel engines. This is a common misconception even today.

chamber was a challenge. And the injection process itself was another point of consternation. He initially attempted to use a technique called solid injection, in which the fuel is driven into the chamber by a plunger or pump, but this proved ineffective. He found that a compressed-air technique worked much better, because it helped atomize the fuel and mix it with the air inside. But, this technique required extra pumps and made the diesel engine heavier and more expensive [42].

Nevertheless, after four years of trial and error, Diesel and his partners debuted their successful engine design. While Diesel calculated that his engine could theoretically reach 75% efficiency, in reality the engine he built clocked in at 26%. Although significantly less than his goal, this was more than double the efficiency of the typical steam engine or other early ICEs of the time [41, 43].

Diesel's engine started selling commercially, and although there were a few bumps in the road, more than 70,000 were in use by 1912, primarily in factories and as electricity generators [43]. In the early 1900s, the diesel engine made its way into military transportation, notably submarines, but because they were especially noisy and spewed clouds of smoke as they ran, it would take until the 1920s for the automotive industry to start adopting them [44, 45].

This left Daimler and Benz as each other's main competitors for the automotive market in Europe at the turn of the 20th century. In 1901, Daimler's company created a car based on one of Maybach's designs. The car had a 35-horsepower engine and maxed out at a speed of over 50 mph [46]. Named "Mercedes" after the daughter of one of Daimler's business partners [38], the car can be considered the first modern automobile in all aspects. Daimler died in 1900 and never saw the completion of this car, but his company continued on. Years later, in 1926, the Daimler and Benz companies merged to form Daimler-Benz AG, and their new automobiles were sold under the now familiar Mercedes-Benz brand.

Other well-known European automobile companies were formed in the late 19th century, including Peugeot S.A. (which originally manufactured coffee grinders and bicycles), Renault S.A., and Fiat S.p.A. In the United States, several individuals were building gasoline-powered vehicles in that same timeframe. Brothers Frank and Charles Duryea were in the bicycle business before creating the first successful American gasoline automobile. Charles designed a gasoline engine-powered carriage, and he and Frank constructed the vehicle. Disagreement over who deserved credit for the invention (as Charles initially designed the car, but Frank claimed to have made significant changes to his brother's design to get the vehicle to work) created a rift between the brothers and a controversy that still lingers today [47, 48]. Nonetheless, the car was successfully driven on the streets in 1893 [49], and the brothers went on to found the Duryea Motor Wagon Company, the first auto manufacturer in America [50].

Ransom E. Olds, whose name would become inextricably linked to the Oldsmobile, was also developing gasoline cars in America around the same time as the Duryea brothers. Olds learned much from working in his father's shop as a machinist and developed a steam engine with a gas burner in the 1880s [51]. After receiving a patent for a gasoline-powered car in 1896, Olds went on to found the Olds Motor Vehicle Company in 1897 [52]. He sold his company, which was renamed Olds Motor Works, and then moved with the company to Detroit. It was then, in 1901, that he designed the Curved Dash Oldsmobile (named for the unique shape of its footboard), which was the first commercially successful American-made automobile.

Olds' factory was also the first to use a modern assembly line process, which enabled the mass-production of the Oldsmobile [52].

The Curved Dash Oldsmobile had a one-cylinder, three-horsepower engine, and was steered with a tiller. Compared to Daimler's 1901 Mercedes, the Oldsmobile technology was somewhat lacking. However, Olds desired to create an affordable car, and his Oldsmobile sold for only $650 (equivalent to about $20k in today's dollars), making it available to middle-class Americans [46].

FIGURE 2.9 1903 Mercedes Simplex.

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FIGURE 2.10 1904 Curved Dash Oldsmobile.

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Competing Technologies

Thus was the state of the automobile industry at the beginning of the 20th century. Horse-drawn carriages were still the primary mode of transportation, but there was ample reason to want a replacement. In the 19th century, horses were the backbone of the economy in North America and Europe—a fact that became all too apparent in the fall of 1872. In an event that came to be known as the Great Epizootic [53], a wave of equine influenza swept through Canada and the United States, bringing city life to a virtual halt as "streetcars[2] stopped running, urban goods stopped moving, and construction sites stopped operating" [54].

In addition to their susceptibility to disease, horses were the cause of much uncleanliness in cities. Bigger cities could have tens or hundreds of thousands of horses, and each horse excreted some 30 pounds of manure and a gallon of urine onto the streets daily—a lot of pollution! Some cities had good systems in place to remove this waste, but in others, it ended up getting dumped into nearby waterways. The waste also attracted flies, many of which carried and spread disease [55].

Given these circumstances, it is no wonder that self-propelled vehicles were gaining in popularity at the turn of the century. In 1900, there were approximately 2,370 cars in New York, Chicago, and Boston combined, about half of which were steam, one-third electric, and one-sixth gasoline [28].

The steam engine was a tried and tested technology, and it was powerful and efficient. But steam was not terribly practical for powering passenger vehicles. Steam engines required a long start-up time to generate enough steam to run. In addition, steam-powered vehicles were limited to about 25–30 miles in range before the water had to be refilled, and they required frequent, laborious maintenance. While steam engines would live on longer in railway locomotives and ships, steam automobiles had all but disappeared by 1920 [28, 56].

Electric and gasoline vehicles were both poised to take the steamers' place at the beginning of the century, and electric cars offered some significant advantages at the time. Gasoline cars were loud, dirty, and produced odorous fumes. Furthermore, gasoline cars had to be started using a hand crank that not only required significant strength to turn but was also potentially dangerous. The crank had a tendency to kick back and caused many a sprained or broken wrist [28]. Changing gears in a gasoline car was also not an easy task. At the time, gasoline cars typically had non-synchronous manual transmissions, which required the driver to time carefully when to shift so that the gears would be rotating at the same speed. This called for a decent amount of skill on the driver's part, and when done incorrectly, the gear teeth would not mesh properly, and the gears would grind [23, 57].

Electric cars, on the other hand, had none of these problems. They were quiet, clean, easy to operate, and more reliable than gasoline cars. Electric vehicles were popular in urban areas, especially with women and professionals like doctors who often drove short distances around the city and made frequent stops [58]. However, there were several drawbacks to EVs, one of the big ones being their limited range. In the early 20th century, the short range was less of an issue, since roads between

[2] For our European readers, streetcars are typically called "trams."

cities in the USA were rough and proved difficult to traverse with any type of automobile.

There were several successful American EV manufacturers during this period, including the Riker Motor Vehicle Company based in New Jersey (Riker would go on to combine with the Pope Manufacturing Company, among other entities, under the Electric Vehicle Company) [28] and the Baker Motor Vehicle Company in Ohio [59]. Walter C. Baker started his company along with his father-in-law and brother-in-law in 1899 and showed his first commercially available two-seater EV (the Imperial Runabout) at America's first auto show in New York City. His car attracted a lot of attention, and by 1906, the Baker Motor Vehicle Company was the top producer of EVs (albeit briefly). Baker gained further celebrity when President William H. Taft's administration purchased a 1909 Baker Electric as one of the White House's first cars (joining the White steamer) [59].

FIGURE 2.11 1912 Baker Electric.

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While the EV enjoyed prominence in the early 20th century, the desire for intercity travel grew as roads began to improve. To lengthen the range of the electric car, batteries needed an upgrade. Thomas Edison was among those who set out to advance the battery, as he believed EVs held a lot of promise. Edison succeeded in creating a battery that would extend the range of EVs up to 100 miles, but his battery was large, heavy, expensive, and hard to maintain [25, 28].

There was one man who succeeded in developing a long-range EV in the early 20th century. Oliver P. Fritchle was a chemist who specialized in smelting before he turned his attention to the electric automotive field. Based in Denver, Colorado, he gained a reputation as a battery repairman for electric vehicles. He began experimenting with his own batteries and soon created a ground-breaking battery that could power a car for 100 miles on a single overnight charge, earning his car the nickname the "100-Mile Fritchle." He sold his first vehicle in 1906 and opened a production plant in Denver in 1908 [60, 61].

FIGURE 2.12 A 1909 ad for a Model A Victoria Phaeton, a.k.a. the 100-Mile Fritchle Electric.

Public Domain Image

Fritchle famously challenged his competitors to an "endurance test" of their cars—a 1,800-mile drive from Lincoln, Nebraska to New York City. No one took him up on his challenge, but he embarked on the excursion anyway. Although the roads were dreadful, Fritchle completed his trek in 20 days of driving with no breakdowns and only one flat tire—an awe-inspiring, triumphant feat at the time [62]. After arriving in New York City, Fritchle decided to go even further and drive to Washington, D.C., bringing the total distance of the trip up to 2,140 miles.

While Fritchle's business was successful, selling hundreds of vehicles in the following years, there was a major drawback to his cars: they were very expensive. Depending on the model, a Fritchle car cost anywhere from $2,000 to $3,600 (approximately $55,000–$100,000 today), making them available only to the wealthy [61].

Another serious hurdle for the 100-Mile Fritchle—and all EVs of the time—was the lack of charging infrastructure. At that time, electricity in the USA was only commonplace in cities, so driving EVs in rural areas was simply not practical. And even within cities, charging presented an issue. In the early 20th century, EV companies and electric utility companies teamed up to try to promote the mutually beneficial goal of expanding battery charging stations. But this proved difficult, and conflicts between the two groups ultimately caused the effort to fail [58].

Perhaps the biggest blow to early EVs, however, was one Henry Ford. While working as an engineer for the Edison Illuminating Company, he spent his free time building a gasoline-powered horseless carriage. He completed his "Quadricycle," equipped with a two-cylinder, four-horsepower engine, in 1896 [63].

He resigned from Edison Illuminating Company in 1899 and formed the Detroit Automobile Company along with several others. The company failed in less than two years. Ford moved on to designing, building, and even driving race cars, in large part to fund his next ventures. In 1901, Ford founded the Henry Ford Company, which he left the following year. After his departure, the company became the Cadillac Motor Car Company [64].

It was then, in 1903, that Ford and several investors created the Ford Motor Company. Their first car was the Model A, followed several years later by the Model N. The Model N was a success and became the best-selling gasoline car in the USA [65]. But it was Ford's Model T that effectively revolutionized the automobile industry.

The Model T, released in 1908, was sturdy and easy to operate, maintain, and repair. Importantly, the Model T was also affordable enough to be within the reach of middle-class Americans. Soon, Ford had more orders than his company could fill. To keep up with demand, Ford implemented modern mass production techniques, including large production plants, standardized parts, and the moving assembly line [63].

At first, the Model T sold for around $850, which was half the price of a typical electric car. The price of the Model T dropped over the following years and demand continued to rise, so that by 1922, half of the cars in America were Model Ts—a truly extraordinary accomplishment [65].

Also in 1922, the Model T's famous nickname was born. A man named Noel Bullock entered his car, which he called "Old Liz," into a race in Pikes Peak, Colorado. Bullock's Model T wasn't painted and is said to have looked like a tin can. As a result, the car was renamed "Tin Lizzie" by spectators. After winning the race, newspapers across the country reported on Tin Lizzie's surprising success, and from that point on, Model Ts were commonly referred to by this nickname [66].

FIGURE 2.13 1919 Ford Model T Sedan.

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In addition to Ford's innovations, another blow to the early electric car was the invention of the electric starter. Charles Kettering, an American electrical engineer, developed a small electric motor powered by a lead-acid battery that could turn over an engine. Then, the battery was recharged by the starter motor, which acted at that point like a generator. The battery could also supply power for lights and spark ignition. Kettering's invention fixed several problems with gasoline vehicles in one go and made the notorious hand crank obsolete [58].

Interestingly, in 1914, Henry Ford and Thomas Edison, who were close friends (what a pair!), announced they were partnering together to develop a long-range, affordable electric car. While the announcement caused a stir in the media, the project ultimately failed after several lackluster prototypes were created.

The sales of electric cars peaked in 1912, with more than 6,000 vehicles being produced by two dozen manufacturers. To compare that with gasoline cars, in the same year over 82,000 Model Ts alone were sold [58]. Roads continued to improve over the next decade, and the long range offered by gasoline cars became increasingly important. Additionally, with the discovery of crude oil in Texas, fuel prices were low enough to be affordable to everyday Americans [23]. By 1935, electric cars had been all but left behind, and ICE-powered vehicles ruled the roads.

A Century of ICE

ICEVs progressed through a variety of phases throughout the 20th century. The 1920s and 1930s brought many technological improvements to automobiles, including hydraulic brakes, automatic transmission, power steering, and closed-body designs. In addition, luxuries, such as heaters and radios became popular. The utilitarian Model T, which had thrust ICE vehicles to the forefront of the consumer market, began to wane in popularity as people opted for more stylish, technologically advanced vehicles.

Toward the end of the 1920s, the automobile market had reached saturation, with demand for replacement cars surging ahead of demand for first-time purchases. To cope with a more static market, a new tactic for selling cars took hold, led in large part by Alfred P. Sloan of General Motors (GM). The idea was to create demand for cars based on stylistic upgrades rather than technological advancement. By annually releasing restyled cars with largely cosmetic changes, GM created an incentive for consumers to trade in their vehicle for a newer model long before the practical life of their vehicle was over. With this mentality, cars started to become larger, faster, and more luxurious.

However, the Great Depression in the United States, and its radiating effects around the world, changed the automotive landscape. With a vastly diminished market, smaller, independent manufacturers were unable to compete with larger companies and many ultimately failed. That left the "Big Three"—GM, Ford, and Chrysler—dominating the American market. With their innovative sales tactics, GM overtook Ford, and by 1936, GM comprised 43% of US automotive sales, followed by Chrysler with 25%, and Ford with 22% [46].

World War II caused another upheaval in the automotive industry. Auto manufacturers stopped manufacturing cars for civilians, and instead churned out military vehicles and a variety of other non-automotive-related military items. Pent-up demand for new cars during the war created a booming market after it ended. The appeal of American-made cars had fallen, however, because they were too large and expensive to operate in countries recovering from the war. European manufacturers quickly filled this gap, and for the first time, the USA began importing cars in significant numbers. At first, many of the imported cars were from British manufacturers, but by the mid-1950s, Volkswagen had made its way into the American market, and its vehicles made up half of all import sales. Given the successes of smaller and mid-sized cars, American manufacturers followed suit, creating a variety of smaller vehicles to complement their fleet of large cars.

Beginning in the late 1950s, new competition arose in the automotive industry from Japan. Japanese manufacturers offered small, high-quality cars that were fuel efficient and boasted contemporary designs. In 1980, Japan rose to the top of the industry, becoming the world's leading auto producer; since 2009, that title has been held by China [67].

While smaller vehicles were popular in the mid-to-late 20th century, larger automobiles, such as the SUV, have significantly gained in popularity in recent times. For example, in 2010 there were about 35 million SUVs in operation worldwide. This number grew to a whopping 200 million in 2019 [68]. Indeed, the automotive industry is experiencing an interesting dichotomy in today's market—supplying the large SUVs and pickup trucks that consumers want, while also pursuing environmentally friendly mobility solutions.

The Last Mile

As we will see throughout this book, we have once again entered a period of competing technologies. The ICE has ruled for the last hundred years, but electric cars are making a comeback. What will the future of passenger cars look like? This is the Big Question. While the battle of propulsion technologies at the turn of the 20th century was driven largely by convenience, practicality, and cost, today's competition is fueled by something else: pollution.

Now, we've actually seen a similar push in the past with the transition from horses to cars. This transition took place over a short amount of time. You can see the stark difference in the photos of the New York Easter parade in Figure 2.14. In 1900, horse-drawn carriages dominated the parade, but by 1913, there was not a horse to be found. The transition happened even faster, really, in just a few years.

What caused this rapid transition? One of the main reasons automobiles replaced horses was pollution. Again, *pollution* was one of the main driving forces behind adopting the automobile. Crazy, right? As mentioned earlier, each horse could excrete tens of pounds of manure daily. Now, multiply that by 100,000 horses in New York City in 1900—that's a lot of horse s***!

With manure-covered streets that festered with flies, odor, and disease, the automobile was clearly the superior choice—the sustainable(!) choice. *But didn't we just trade one polluting propulsion system for another?* This is an important lesson, and it's a major theme throughout this book. The car got rid of the visible (and odorous) horse pollution and eliminated its negative impacts. If we only cared about pollution on the streets, we could even say that the automobile was a "zero manure vehicle!" However, as car use increased and more miles were driven (3.2 trillion miles were driven in 2018 in the USA alone), an unintended consequence came along for the ride: vehicle emissions. They're not as visible as horse manure, but they're damaging in their own right—damaging in a different way. So, what emissions did we trade manure for and how are they formed? Let's take a look in the next chapter.

FIGURE 2.14 The New York Easter parade in 1900 (top) and 1913 (bottom)—spot the difference!

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